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Practical 5 Part II

What is a Join?

A **JOIN** combines records from two or more tables using a related column.

Types of Joins:

1. **INNER JOIN** → Returns only matching records.
2. **LEFT JOIN** → Returns all records from the left table and matching records from the right table.
3. **RIGHT JOIN** → Returns all records from the right table and matching records from the left table.
4. **FULL OUTER JOIN** → Returns all records from both tables (not available in MySQL).
5. **CROSS JOIN** → Returns the Cartesian product of both tables.
6. **SELF JOIN** → Joins a table to itself.

1. Customer Table

Column	Data Type	Constraints
customer_id	NUMBER (PK)	PRIMARY KEY, AUTO-INCREMENT

name	VARCHAR2(100)	NOT NULL
email	VARCHAR2(100)	UNIQUE
phone	VARCHAR2(15)	NOT NULL
address	VARCHAR2(255)	NULLABLE

2. Product Table

Column	Data Type	Constraints
product_id	NUMBER (PK)	PRIMARY KEY
name	VARCHAR2(100)	NOT NULL
category	VARCHAR2(50)	NOT NULL
price	DECIMAL(10,2)	NOT NULL
stock_quantity	INT	NOT NULL

3. Order_Details Table

Column	Data Type	Constraints
order_id	NUMBER (PK)	PRIMARY KEY
customer_id	NUMBER (FK)	FOREIGN KEY REFERENCES Customer(customer_id)
order_date	DATE	NOT NULL
total_amount	DECIMAL(10,2)	NOT NULL

4. Order_Item Table

Column	Data Type	Constraints

order_id	NUMBER (FK)	FOREIGN KEY REFERENCES Order_Details(order_id)
product_id	NUMBER (FK)	FOREIGN KEY REFERENCES Product(product_id)
quantity	INT	NOT NULL
subtotal	DECIMAL(10,2)	NOT NULL

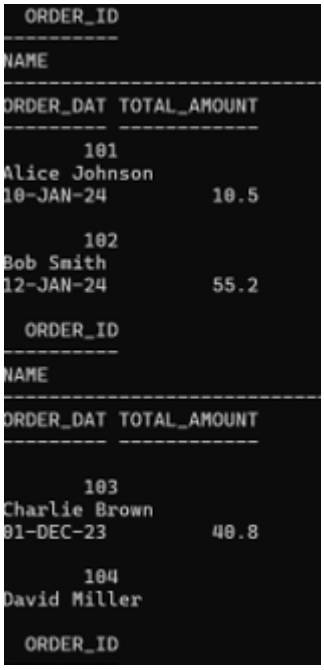
5. Employee Table

Column	Data Type	Constraints
employee_id	NUMBER (PK)	PRIMARY KEY
name	VARCHAR2(100)	NOT NULL
role	VARCHAR2(50)	NOT NULL
salary	DECIMAL(10,2)	NOT NULL
hire_date	DATE	NOT NULL

Examples of Joins

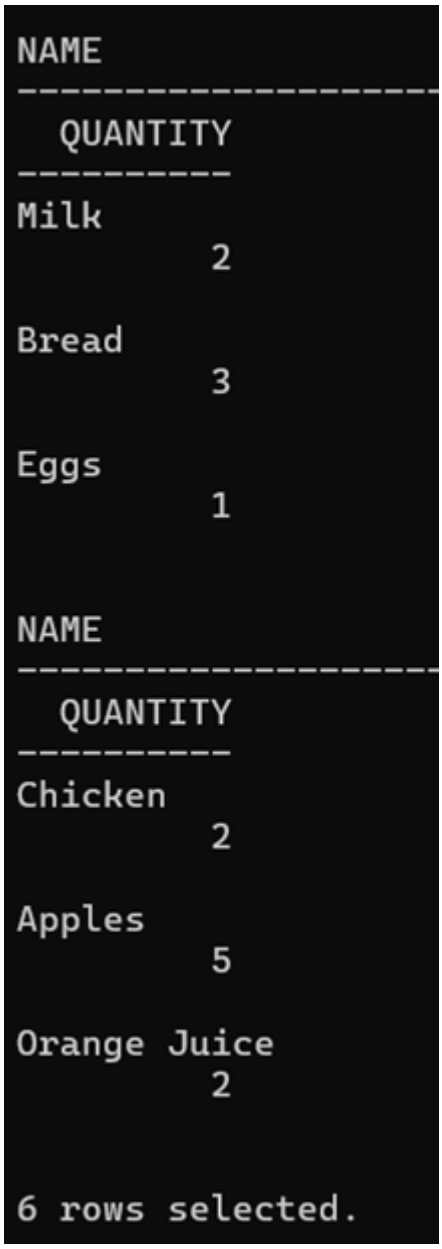
INNER JOIN: Get order details with customer names

```
SELECT o.order_id, c.name, o.order_date, o.total_amount
FROM Order_Details o
INNER JOIN Customer c ON o.customer_id = c.customer_id;
```



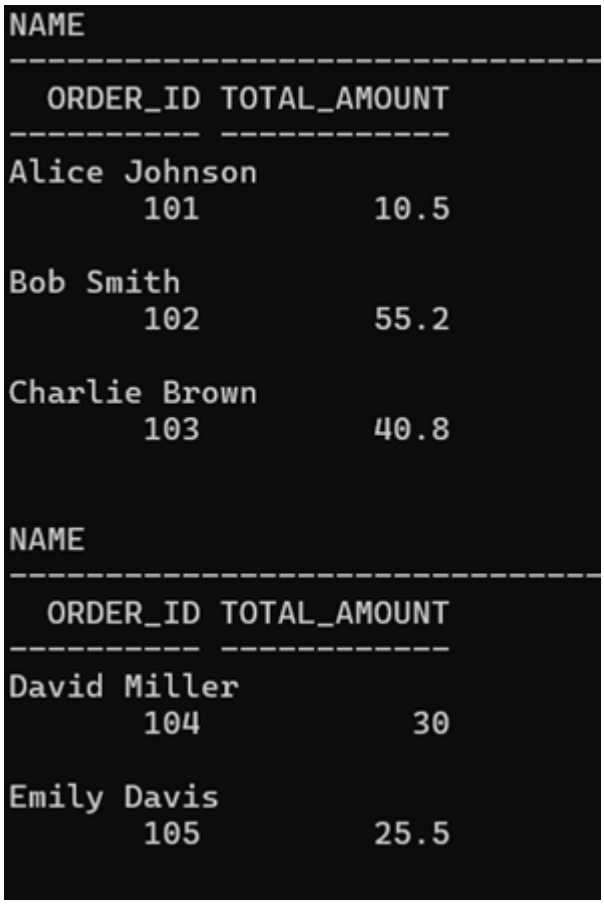
INNER JOIN: Retrieve product names and their order quantities

```
SELECT p.name, oi.quantity
FROM Order_Item oi
INNER JOIN Product p ON oi.product_id = p.product_id;
```



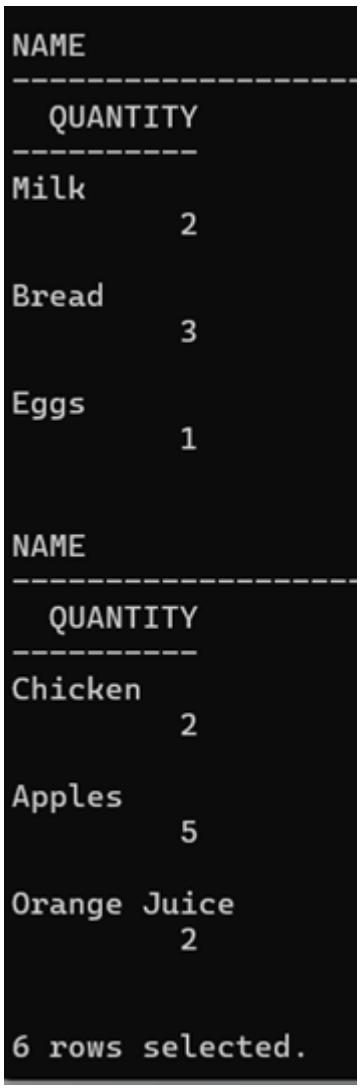
LEFT JOIN: Get all customers and their orders (including those who never ordered)

```
SELECT c.name, o.order_id, o.total_amount
FROM Customer c
LEFT JOIN Order_Details o ON c.customer_id = o.customer_id;
```



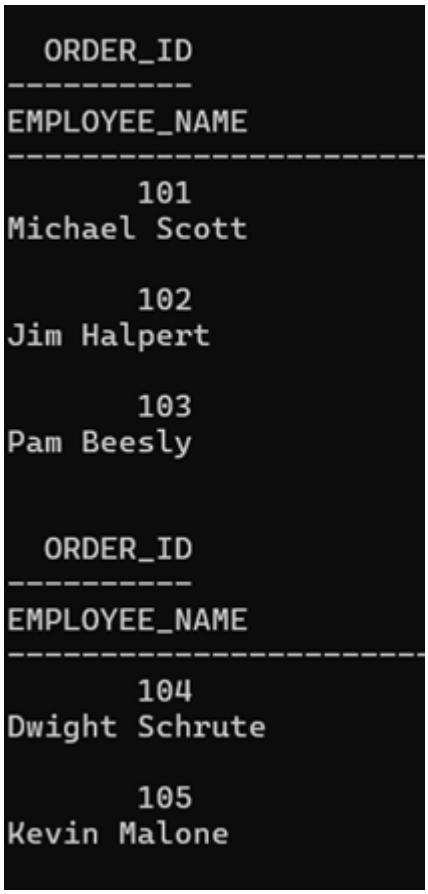
LEFT JOIN: Retrieve all products and their order details (including those not ordered yet)

```
SELECT p.name, oi.quantity
FROM Product p
LEFT JOIN Order_Item oi ON p.product_id = oi.product_id;
```



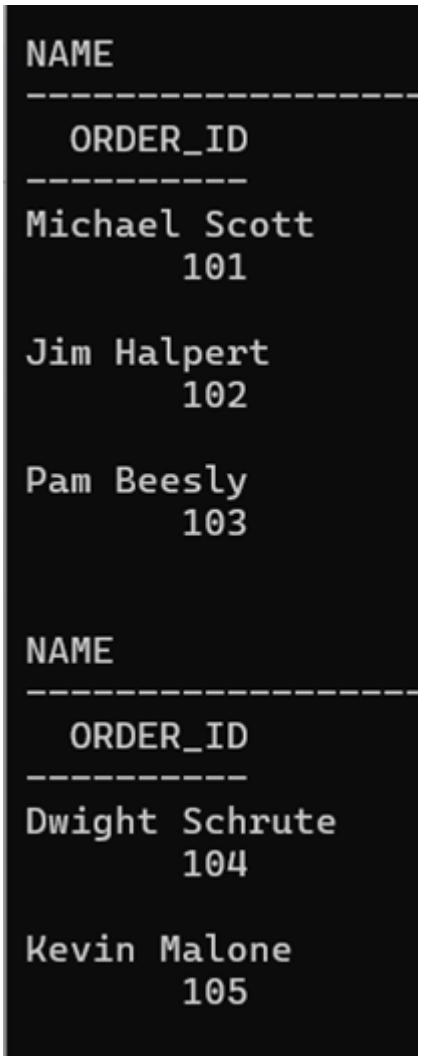
RIGHT JOIN: Get all orders with or without employee assigned

```
SELECT o.order_id, e.name AS employee_name
FROM Order_Details o
RIGHT JOIN Employee e ON o.customer_id = e.employee_id;
```



RIGHT JOIN: Retrieve employees who processed orders

```
SELECT e.name, o.order_id
FROM Employee e
RIGHT JOIN Order_Details o ON e.employee_id = o.customer_id;
```



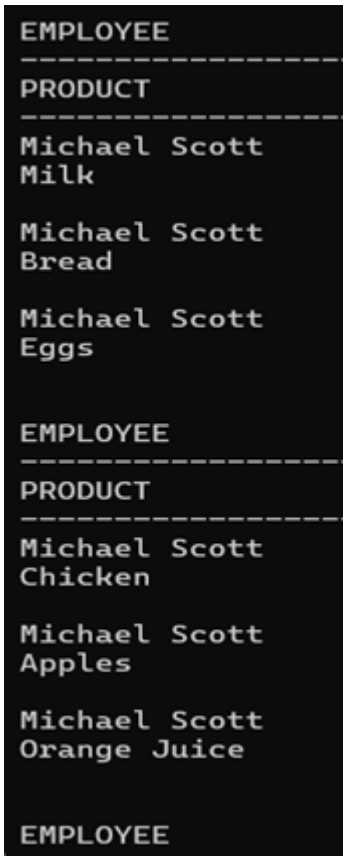
FULL OUTER JOIN: Get all customers and orders (Oracle SQL only)

```
SELECT c.name, o.order_id, o.total_amount
FROM Customer c
FULL OUTER JOIN Order_Details o ON c.customer_id =
o.customer_id;
```

```
NAME
-------------------------
  ORDER_ID TOTAL_AMOUNT
---------- ------------
Alice Johnson
       101         10.5

Bob Smith
       102         55.2

Charlie Brown
       103         40.8


NAME
-------------------------
  ORDER_ID TOTAL_AMOUNT
---------- ------------
David Miller
       104           30

Emily Davis
       105         25.5
```

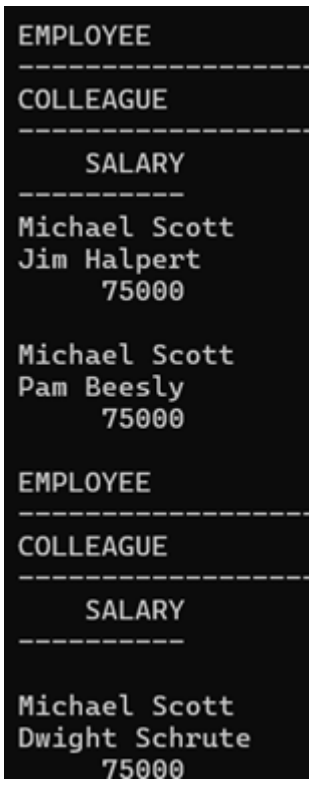
CROSS JOIN: Show all possible employee-product assignments

```
SELECT e.name AS employee, p.name AS product
FROM Employee e
CROSS JOIN Product p;
```



SELF JOIN: Find employees earning more than their colleagues

```
SELECT e1.name AS Employee, e2.name AS Colleague, e1.salary FROM Employee e1
JOIN Employee e2 ON e1.salary > e2.salary;
```



SELF JOIN: Find employees working under the same manager

```
SELECT e1.name AS Employee, e2.name AS Manager
FROM Employee e1
JOIN Employee e2 ON e1.role = 'Cashier' AND e2.role =
'Manager';
```

Joins Tasks

1. Retrieve **customer names** along with their orders.

```
SQL> SELECT C.name AS Customer_Name, O.order_id, O.order_date
  2  FROM CUSTOMER C
  3  JOIN ORDER_DETAILS O ON C.customer_id = O.customer_id;

CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
Alice Johnson
       101 10-JAN-24

Bob Smith
       102 12-JAN-24

Charlie Brown
       103 01-DEC-23


CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
David Miller
       104 05-NOV-23

Emily Davis
       105 10-FEB-24
```

2. Show **product names** and their **order quantities**.

```
SQL> SELECT P.name AS Product_Name, OI.quantity
  2  FROM PRODUCT P
  3  JOIN ORDER_ITEM OI ON P.product_id = OI.product_id;

PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY
----------
Milk
         2

Bread
         3

Eggs
         1


PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY
----------
Chicken
         2

Apples
         5

Orange Juice
```

3. List all customers and their orders (**including those who never ordered**).

```
SQL> SELECT C.name AS Customer_Name, O.order_id, O.order_date, O.total_amount
  2  FROM CUSTOMER C
  3  LEFT JOIN ORDER_DETAILS O ON C.customer_id = O.customer_id;

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
Alice Johnson
       101 10-JAN-24         10.5

Bob Smith
       102 12-JAN-24         55.2

Charlie Brown
       103 01-DEC-23         40.8


CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
David Miller
       104 05-NOV-23           30

Emily Davis
       105 10-FEB-24         25.5
```

4. Retrieve **all products and their order details** (**including those not ordered yet**).

```
SQL> SELECT P.name AS Product_Name, OI.order_id, OI.quantity, OI.subtotal
  2  FROM PRODUCT P
  3  LEFT JOIN ORDER_ITEM OI ON P.product_id = OI.product_id;

PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY   SUBTOTAL
---------- ---------- ----------
Milk
       101          2          5

Bread
       101          3        5.4

Eggs
       102          1        3.2


PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY   SUBTOTAL
---------- ---------- ----------
Chicken
       102          2         15

Apples
       103          5          6

Orange Juice
```

5. Find employees who have **processed orders**.

```
SQL> SELECT DISTINCT E.name AS Employee_Name, E.role
  2  FROM EMPLOYEE E
  3  JOIN ORDER_DETAILS O ON E.employee_id = E.employee_id;

EMPLOYEE_NAME
--------------------------------------------------------------------
ROLE
--------------------------------------------------------------------
Michael Scott
Manager

Jim Halpert
Cashier

Pam Beesly
Sales Associate


EMPLOYEE_NAME
--------------------------------------------------------------------
ROLE
--------------------------------------------------------------------
Dwight Schrute
Supervisor

Kevin Malone
Cashier
```